

# Effects of Two Perforation Designs on Monopile Structures for Offshore Wind Turbines

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## Introduction

Offshore wind turbines are commonly supported by cylindrical steel structures, known as monopiles (Garcia 2012), that are subjected to harsh environments that encourage corrosion and biofouling. Maher (2018) proved that cathodic, perforated designs protected the structure from corrosion and created a habitat for marine life. The goal of this project is to determine how perforations will affect the integrity of life-sized monopiles.

## Objectives

- To determine existing forces on solid monopiles.
- To design and model alternative perforated monopiles to compare stress and strain values.

## Methods

### Determine Forces that Exist on a Solid Monopile

- Find existing research and similar experiments.
- Calculate the horizontal forces acting on a solid monopile from seabed to surface.

### Design and Model Different Perforated Monopiles

- Generate a CAD model of a solid monopile.
- Generate a CAD model of a perforated monopile with three equally spaced columns of two holes.
- Generate a CAD model of a perforated monopile with a spiral hole pattern.
- Use Fusion 360® software to analyze the stress and strain concentrations on each monopile once existing forces are applied.
- Adjust the diameter, location, and number of holes as necessary to produce a stable structure.

## Results

### Determine Forces that Exist on a Solid Monopile

Horizontal wave forces were calculated using Equation 1: Morison's Equation.

$$F_H = C_m \rho V \dot{u} + \frac{1}{2} \rho A u |u|$$

Equation 1: Morison's Equation

Figure 1 shows the profile for the drag and inertial wave forces acting along the height of the monopile in 1-m intervals. Forces are shown to be weaker at the seabed and increase logarithmically as they approach the surface.

## Results (cont.)

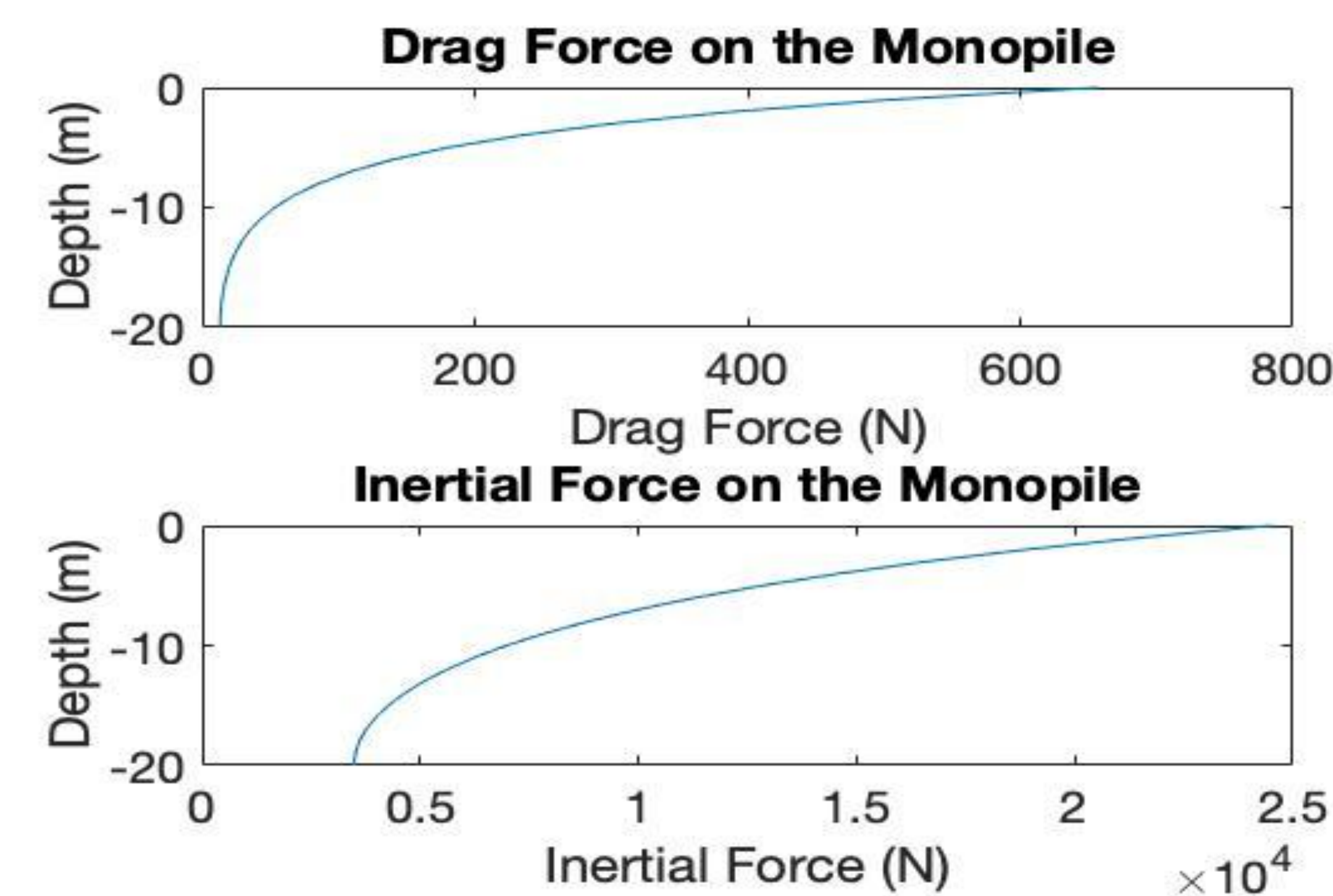


Figure 1: Drag Force and Inertial Force Profiles

Figure 2 shows the profiles for the hydrostatic and the dynamic pressures. Pressure values were calculated at points along the structure in 1-m intervals. Unlike the wave forces, the pressures increase with depth.

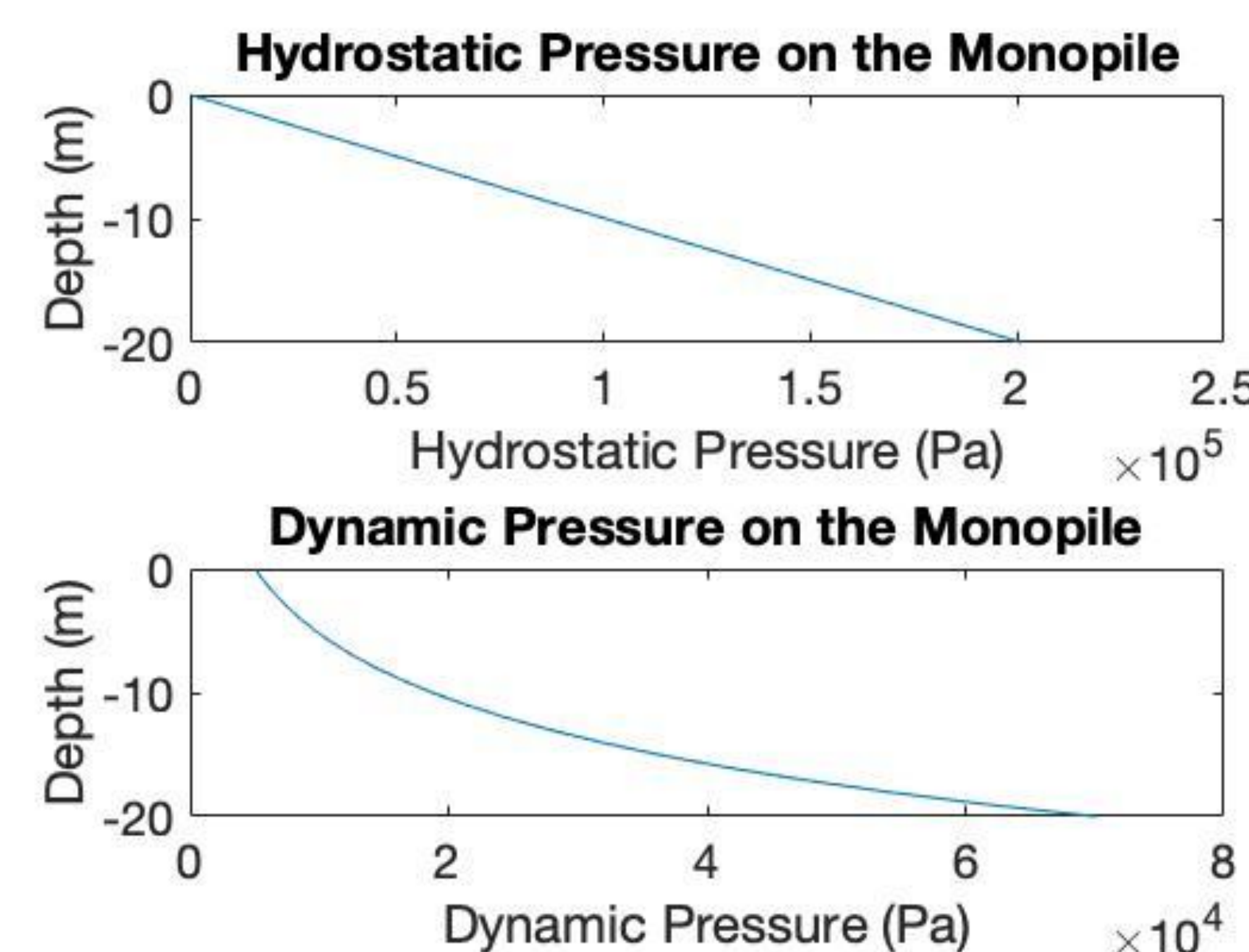


Figure 2: Hydrostatic and Dynamic Pressure Profiles

### Design and Model Different Perforated Monopiles

Figure 3 shows the three tested CAD designs. Design (a) is a solid monopile. Design (b) is a perforated monopile with two levels of 3 holes each. Design (c) is a perforated monopile with a spiral design.

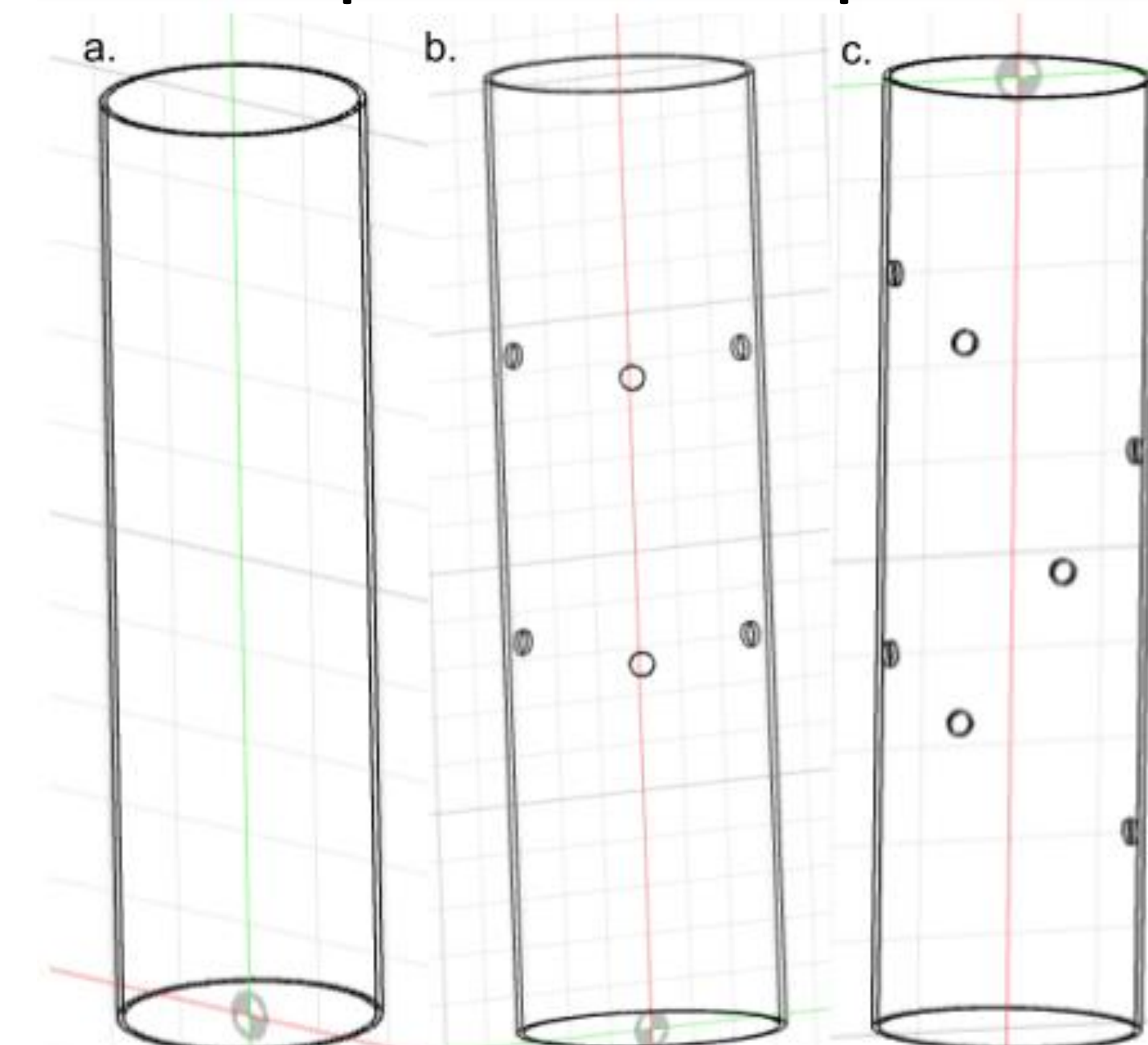


Figure 3: Monopile CAD designs

## Results (cont.)

Figures 4 & 5 show the results obtained for stress & strain on the 3 designs using Fusion 360®.

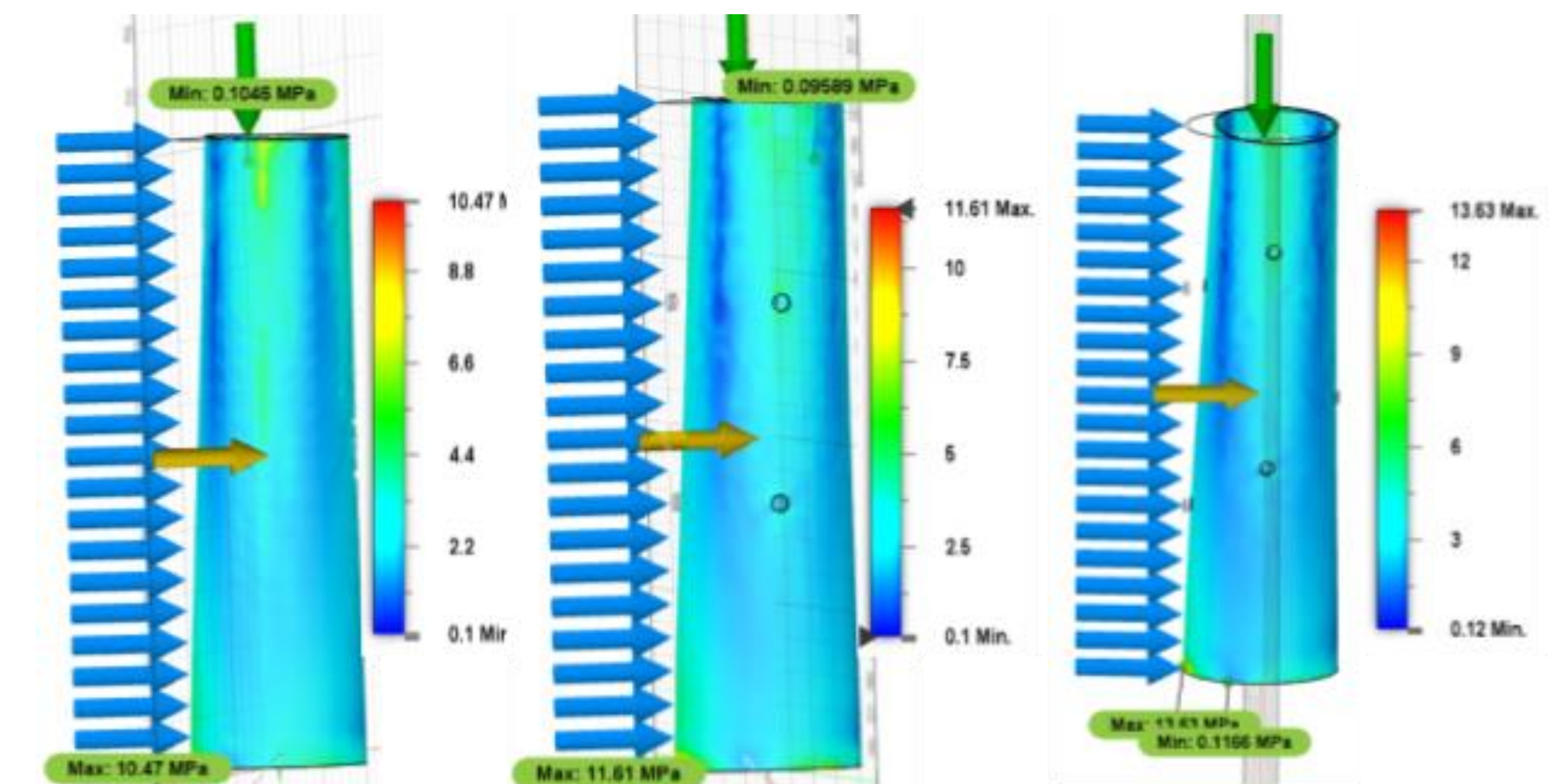


Figure 4: Comparison of Stress on Monopile Designs

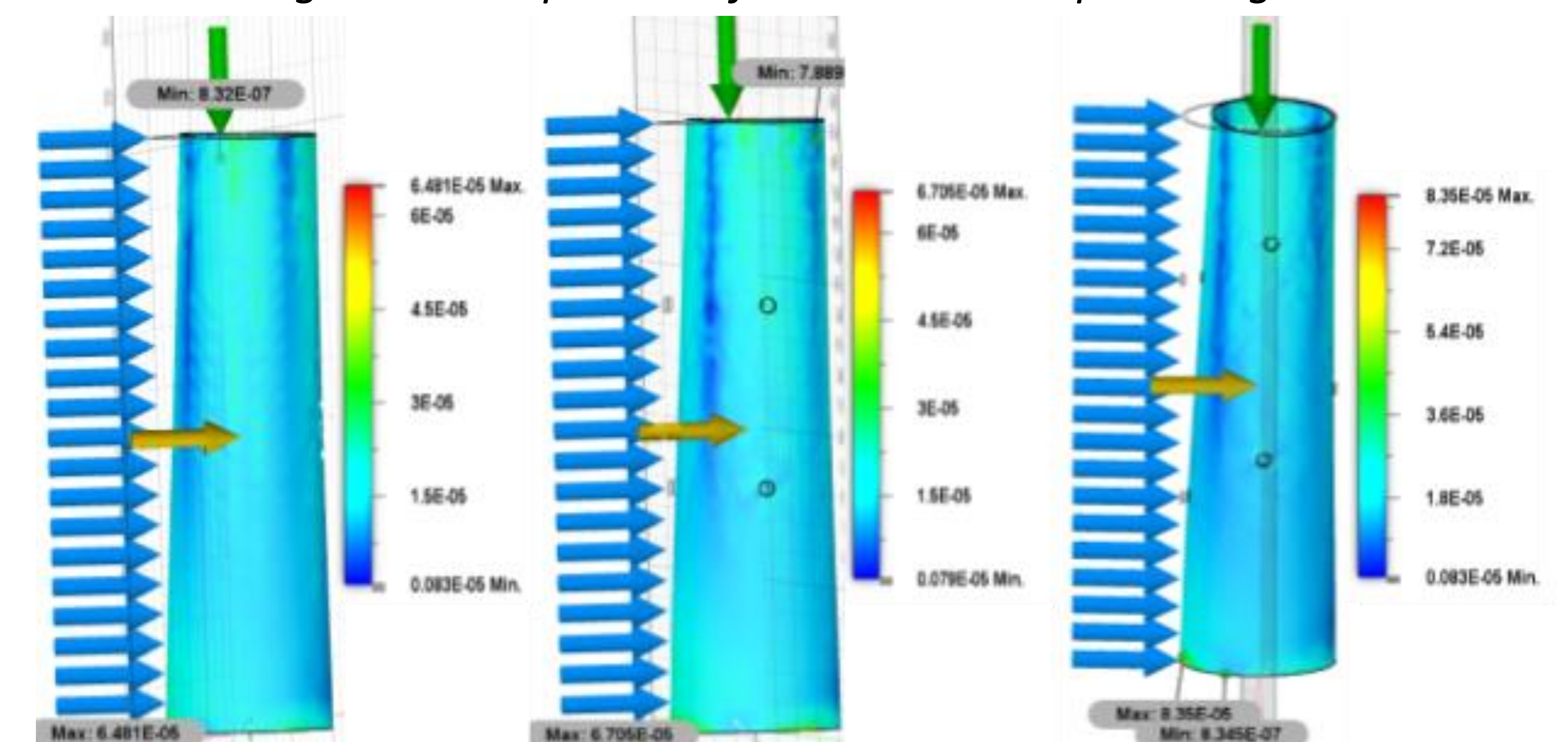


Figure 5: Comparison of Strain on Monopile Designs

## Discussion/Conclusion

For 0.5-m diameter perforations, maximum stresses & strains were far below the tensile strength & allowable elongation of A572Gr50 Steel used in the industry. Design B had smaller effects than Design C, both were within allowable limits. Perforations for life-sized models are promising, more details [here](#). Next steps:

- Incorporate wind loads into the designs.
- Generate a range of feasible hole sizes & patterns.
- Analyze water circulation inside monopiles.
- Study habitat potential for larval settlement.

## References

- Maher, M. M., and G. Swain. "The Corrosion and Biofouling Characteristics of Sealed vs. Perforated Offshore Monopile Interiors..." *OCEANS 2018 MTS/IEEE Charleston*, 2018.
- Garcia, C. G. "Design & Calculus of the Foundation Structure of an Offshore Monopile Wind Turbine." *Facultat Nautica Barcelona*, 2012.

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